

Do we need a stressor to be stressed? Insights from cardiac regulation¹

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Abstract: In consideration of existing contradictory findings, the aim of the present study was to overcome the limitations of traditional cardiac reactivity studies. First, we examined the cardiac activation that takes place in response to ruminating about stress, independently of the presence of a real stressor, and second, we focused on the underlying physiological components of cardiac regulation during rumination, that is, the sympathetic and parasympathetic components. In 60 participants (27 men and 33 women; mean age = 33.4 ± 9.5 years), we recorded cardiac impedance and heart rate during baseline, an Anger Recall Interview, a reading task, and two recovery periods. Rumination was experimentally manipulated by the use of a distracter. Nondistracted participants showed higher sympathetic activity as suggested by shorter pre-ejection period, higher low-frequency heart rate variability (LF-HRV), higher ratio of LF to high frequency HRV (HF-HRV), and diminished parasympathetic activity (lower HF-HRV). The results underscore the association of rumination with a series of well-known risk factors for health.

Key words: rumination, sympathetic nervous system, parasympathetic nervous system, autonomic balance, risk factors.

The fact that the presence of stressful events may lead to the development and maintenance of somatic disorders is now recognized and accepted (Krantz & McCeney, 2002). Until a few decades ago, this process has been attributed to an excessive response of the individual to stressful events (i.e. physiological hyper-reactivity). The “reactivity hypothesis” posits that the repeated activation of the stress response over time can result in several risk factors for cardiovascular diseases (Schwartz, Gerin, Davidson, Pickering, Brosschot, Thayer, Christenfeld, & Linden, 2003). A recent meta-analysis of prospective evidence (Chida & Steptoe, 2010) showed that greater reactivity

to stress is associated over time with clinical cardiovascular events; however, debates continue about the conclusiveness of the evidence. Moreover, there is increasing evidence of blunted reactivity to stressors as a health risk factor (Phillips, 2011; Phillips, Der, & Carroll, 2009).

In previous studies, we have highlighted two major limitations of past reactivity studies that may have restricted our ability to detect laboratory-to-life generalizability: (a) the focus on blood pressure (BP) or heart rate (HR) reactivity per se (Ottaviani, Shapiro, Davydov, & Goldstein, 2008; Ottaviani, Shapiro, Goldstein, James, & Weiss, 2006); and (b) the

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restricted emphasis on the physiological activation that takes place during a brief laboratory stressor (Ottaviani, Shapiro, Davydov, Goldstein, & Mills, 2009; Ottaviani, Shapiro, Goldstein, & Mills, 2007). With regard to the first issue, we demonstrated that the use of the underlying physiological components of HR and BP response significantly increases the ability of reactivity studies to predict cardiovascular changes in daily life (Ottaviani et al., 2006; 2007; 2008). As to the second limitation, there is now increasing evidence that prolonged physiological activity predicts pathogenic states and organic disease far beyond the immediate cardiovascular reaction (Heponiemi, Elovainio, Pulkki, Puttonen, Raitakari, & Keltikangas-Järvinen, 2007). It is now considered adaptive to react when we have to face a stressful event, and hyper-reactivity becomes maladaptive if it continues when the source of stress is no longer present (Pieper, Brosschot, van der Leeden, & Thayer, 2010).

The aim of the present study was to overcome both of those limitations by studying cardiac activity during periods in which the source of stress is no longer present (i.e. rumination) and focusing on its autonomic nervous system components (i.e. sympathetic and parasympathetic).

According to the guidelines (Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology, 1996), the high-frequency component of HR variability (HF-HRV) reflects parasympathetic activity, and the low-frequency component of HRV (LF-HRV) is proportional to sympathetic activity but influenced by parasympathetic tone. The interpretation of LF-HRV as primarily an index of sympathetic tone has been commonly derived by the calculation of the ratio of LF- to HF-HRV. For example, Montano and colleagues (Montano, Ruscone, Porta, Lombardi, Pagani, & Malliani, 1994) demonstrated that LF/HF-HRV correlates with the tilt angle of a series of passive head-up tilt steps, a commonly used method to elicit different degrees of sympathetic activation.

Considering that the sympathovagal balance concept has been criticized (Berntson, Bigger,

Eckberg, Grossman, Kaufmann, Malik, Nagaraja, Porges, Saul, Stone, & van der Molen, 1997; Eckberg, 1997), and in light of the fact that it captures a different aspect of the same physiological phenomenon (Goedhart, Willemssen, Houtveen, Boomsma, & De Geus, 2008), the pre-ejection period (PEP) was also used as an index of sympathetic cardiac control (Berntson, Lozano, Chen, & Cacioppo, 2004).

Reduced HRV has been associated on the one hand with an increased risk of hypertension, cardiovascular disease, and mortality in general (see Thayer & Lane, 2007 for a review), and on the other with the presence of various forms of psychopathology such as anxiety and depression (Friedman & Thayer, 1998; Rottenberg, Salomon, Gross, & Gotlib, 2005), often characterized by a marked presence of chronic rumination (Nolen-Hoeksema, Wisco, & Lyubomirsky, 2008).

A tool that has been proven to be particularly suitable to evaluate the accompanying physiological effects of a stressful event long after its occurrence is the Anger Recall Interview (Ironson, Taylor, Boltwood, Bartzokis, Dennis, Chesney, Spitzer, & Segall, 1992). If participants are not distracted right after, this task is particularly effective in evoking prolonged and sustained physiological activity, hypothetically due to the ruminative thoughts associated with the event that they recalled (see Brosschot, Gerin, & Thayer, 2006 for a review). Rumination has been recently proposed as a major factor in the relationship between cardiovascular disease and psychiatric comorbidity (Larsen & Christenfeld, 2009). We previously hypothesized that the negative effects of rumination on health could be attributed to the autonomic dysregulation that characterizes this mental process. In agreement with a number of studies (Key, Campbell, Bacon, & Gerin, 2008; Verkuil, Brosschot, de Beurs, & Thayer, 2009), we found HRV reduction during rumination, as measured by both low HF-HRV and baroreflex sensitivity (BRS) (Ottaviani et al., 2009).

In the present study, we used the Anger Recall Interview, coupled with a distracter condition, to evoke rumination in the laboratory

and study its sympathetic/parasympathetic correlates.

Methods

Sample

Participants were recruited by announcements, and exclusionary criteria were psychiatric disorders, significant health problems, use of drugs or medications that might affect cardiovascular function and/or catecholamines, obesity (body mass index $> 32 \text{ kg/m}^2$), diagnosis of hypertension, menopause, use of oral contraceptives during the previous 6 months, and pregnancy or childbirth within the last 12 months. The sample was composed of 27 men (mean age = 31.8 ± 10.1 years) and 33 women (mean age = 34.7 ± 8.6 years) and 12 Asian, 23 Caucasian, 17 African, and 8 Latin-Americans. Participants were paid US\$100. The protocol was approved by the University of California, Los Angeles, Institutional Review Board.

Procedure

All experimental sessions took place between 7:00 a.m. and 12:00 p.m. Participants were asked to refrain from (a) drinking alcohol, tea, or coffee, and (b) strenuous exercise the morning of testing. After providing written informed consent, participants were seated in a comfortable chair in the experiment room and instrumented for cardiovascular monitoring. The experimental protocol consisted of an initial 10-min baseline period, followed by a 5-min reading task, a 10-min rest period, and a 5-min anger recall interview. The reading task required participants to read aloud a passage about the ocean using "a normal tone of voice and normal rate of speech." To insure minimal emotional arousal, participants were instructed they would not be evaluated for reading style or comprehension. The experimental session ended with a 10-min resting period. During the first 2 min of this resting period, half the sample was randomly assigned to the distraction condition, in which the laboratory phone rang, the experimenter went into an adjacent room

leaving the door open, and spoke loudly about job issues for approximately 2 min. For the non-distracted subjects the experimenter left the room and closed the door.

Psychophysiological assessment

Electrocardiogram (ECG) was monitored with a multitrace recorder (AcqKnowledge; Biopac System, Santa Barbara, CA, USA) and a standard electrode configuration (right clavicle and precordial site V6). Three disposable Ag-AgCl electrodes (ConMed Corp.) were used. The signal was digitized at 2000 Hz, and the intervals between two consecutive R waves of the ECG were derived using the AcqKnowledge software. Each epoch was manually checked and corrected for artifacts.

Heart-rate variability-related measures were obtained using the HRV Analysis Software (Niskanen, Tarvainen, Ranta-Aho, & Karjalainen, 2004). The LF and HF bands were chosen according to the guidelines (Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology, 1996).

A Minnesota Impedance Cardiograph (ICG; Bio-Impedance Technology, Chapel Hill, NC, USA) measured the impedance signal according to the published guidelines using four bands of disposable cardiograph electrode tape (Label Technologies, Inc., Chapel Hill, NC, USA) placed circumferentially around the neck, chest, and abdomen (Sherwood, Allen, Fahrenberg, Kelsey, Lovallo, & van Doornen, 1990). Measurements were taken to ensure that the electrodes were at the appropriate distance from one another. Each signal was continuously recorded in 30-s epochs during baseline, each stressor task, and recovery. The Cardiac Output Program (Bio-Impedance Technology, Chapel Hill, NC, USA) acquired, stored, and processed the impedance cardiographic signal via a PC. Each epoch was verified by visual inspection to ensure the accuracy of the event waveform scoring. The pre-ejection period (PEP) was derived from the ensemble-averaged waveforms.

Respiration was recorded with the Biopac Respiratory Effort Transducer (Biopac System,

Santa Barbara, CA, USA), a belt placed around the lower rib cage measuring changes in chest circumference.

The ECG, ICG, and respiration were continuously recorded during baseline, each stressor task, and rest periods.

Data analyses

All data are expressed as means (*SD*). Differences at $p < .05$ were regarded as significant. Laboratory data processing and data analyses were performed with the software modules of Systat 9.0 (Systat Software Inc., Richmond, California, USA). LF-HRV and HF-HRV are expressed in normalized units (n.u.) in the analyses.

Reactivity change scores (Δ) were computed by subtracting the initial baseline average value from each average task value. Recovery scores were determined by subtracting the mean level obtained during the baseline from the average level measured during the recovery period after each task. Averages for baseline, tasks, and recovery were computed for each entire time period (10, 5, and 10 min, respectively).

Sex differences were analyzed using *t*-tests.

Traditional HR reactivity results, as well as the appropriate analyses to control for the presence of pre-existing differences between the distraction and the nondistraction subgroups, have been described elsewhere (Ottaviani, Shapiro, & Fitzgerald, 2011). In accordance with the specific aims of the present study, baseline levels of PEP, HF-HRV, LF-HRV, and LF/HF-HRV were compared using *t*-tests.

As respiration amplitude and depth can influence HRV measures, paired *t*-tests comparing respiratory rates during baseline and tasks were used to see whether changes in respiration rate affected HRV-related variables.

To determine the effectiveness of the anger recall interview in inducing psychophysiological activation compared with baseline and the neutral reading task that acted as a control for vocalization, analyses of variance (ANOVAs) with task as a repeated measure (baseline, anger recall, reading) were conducted for PEP, HF-HRV, LF-HRV, and LF/HF-HRV. To evaluate the effectiveness of the distracter condition

in determining greater recovery to baseline levels after the anger recall task, *t*-tests by group were performed on change scores for each physiological variable.

Results

The sociodemographic characteristics of the sample as well as data on the efficacy of the distracter in determining the presence/absence of self-reported state rumination during recovery from the Anger Recall Interview have been described elsewhere (Ottaviani et al., 2011).

Baseline differences between the distracted and the nondistracted groups were not significant for the examined variables (see Table 1).

With regard to sex, women showed a stronger baseline sympathetic activation compared to men, as suggested by the difference in LF-HRV n.u., 61.9 (6.3) versus 58.4 (5.9); $t(2, 58) = -2.23$; $p = .03$, and PEP, 111.3 (12) versus 120.7 (18.3); $t(2, 58) = 2.28$; $p = .03$. No other sex differences emerged.

Respiratory rate was virtually identical between baseline and each task condition, and did not differ between tasks.

The repeated-measures ANOVAs revealed a significant effect of task for: PEP, $F(2, 118) = 21.06$; $p < .0001$; HF-HRV n.u., $F(2, 118) = 238.58$; $p < .0001$; LF-HRV n.u., $F(2, 118) = 163.85$; $p < .0001$; and LF/HF-HRV, $F(2, 118) = 146.08$; $p < .0001$. Pairwise *t*-tests indicated that the anger interview led to greater activation, (a) compared with baseline for: PEP, $t(1, 59) = 5.26$; $p < .0001$; HF-HRV n.u., $t(1, 59) = 15.37$; $p < .0001$; LF-HRV n.u., $t(1, 59) = -14.02$; $p < .0001$; LF/HF-HRV, $t(1, 59) = -12.23$; $p < .0001$; and (b) compared with the reading task for: PEP, $t(1, 59) = 5.72$; $p < .0001$; HF-HRV n.u., $t(1, 59) = 16.56$; $p < .0001$; LF-HRV n.u., $t(1, 59) = -12.29$; $p < .0001$; LF/HF-HRV, $t(1, 59) = -12.00$; $p < .0001$. Specifically, PEP and HF-HRV were lower while LF-HRV n.u. and LF/HF-HRV were higher during the Anger Interview compared with both baseline and the reading task.

No significant differences between baseline and reading emerged, except for the increase in

Table 1 Physiological variables during baseline, the Anger Recall Interview, the reading task and recovery periods for distracted and non distracted participants (mean and *SD*)

		Distracted (<i>n</i> = 30)	Nondistracted (<i>n</i> = 30)	<i>p</i>
Baseline	PEP (ms)	115.9 (16.5)	116.9 (16.4)	.79
	HF-HRV (n.u.)	32.1 (3.8)	32.0 (4.7)	.94
	LF-HRV (n.u.)	61.4 (6.4)	58.7 (6.0)	.10
	LF/HF-HRV (ms ²)	1.9 (.4)	1.9 (.4)	.46
Anger recall (Δ)	PEP (ms)	-6.4 (7.5)	-5.9 (10.1)	.82
	HF-HRV (n.u.)	-10.8 (5.9)	-11.1 (5.1)	.86
	LF-HRV (n.u.)	14.8 (7.9)	13.8 (7.9)	.66
	LF/HF-HRV (ms ²)	2.0 (1.4)	1.9 (1.0)	.76
Recovery Anger Interview (Δ)	PEP (ms)	-2.34 (9.8)	-3.64 (8.8)	.60
	HF-HRV (n.u.)	-1.97 (5.0)	-8.66 (5.4)	< .0001
	LF-HRV (n.u.)	.48 (5.7)	11.78 (7.8)	< .0001
	LF/HF-HRV (ms ²)	.29 (1.0)	1.36 (.9)	< .0001
Reading (Δ)	PEP (ms)	-2.04 (6.7)	-.02 (7.5)	.30
	HF-HRV (n.u.)	.29 (1.9)	-.26 (1.9)	.26
	LF-HRV (n.u.)	1.01 (2.5)	.94 (2.9)	.92
	LF/HF-HRV (ms ²)	.02 (.1)	.05 (.1)	.37
Recovery Reading (Δ)	PEP (ms)	.07 (5.4)	1.39 (7.7)	.48
	HF-HRV (n.u.)	1.64 (2.4)	1.91 (2.8)	.68
	LF-HRV (n.u.)	-2.19 (2.8)	-2.11 (3.9)	.93
	LF/HF-HRV (ms ²)	-.17 (.2)	-.15 (.2)	.82

Note. HF-HRV = high-frequency heart rate variability; LF/HF-HRV = ratio of low-frequency to high-frequency heart rate variability; LF-HRV = low-frequency heart rate variability; PEP = pre-ejection period.

both LF-HRV *n.u.*, $t(1, 59) = -2.81$; $p = .01$, and LF/HF-HRV, $t(1, 59) = -2.10$; $p = .04$, during the task. Considering the significant effects of reading aloud on these sympathetic measures, two repeated-measure ANOVAs (baseline, Anger Recall) were performed on LF-HRV *n.u.* and LF/HF-HRV using change scores during the reading task as a covariate in the analyses. The effect of task remained significant for both LF-HRV *n.u.*, $F(1, 58) = 171.76$; $p < .0001$, and LF/HF-HRV, $F(1, 58) = 135.08$; $p < .0001$, thus demonstrating that the Anger Recall is associated with a sympathetic activation that goes far beyond the effects of vocalization alone.

The “distracter” condition was effective in determining greater recovery to baseline values (smaller change scores) compared to the “non-distracter” condition (Table 1) for HF-HRV *n.u.*, $t(2, 58) = -4.96$; $p < .0001$; LF-HRV *n.u.*, $t(2, 58) = 6.40$; $p < .0001$; and LF/HF-HRV, $t(2, 58) = 4.30$; $p < .0001$, but not for PEP ($p = .60$). Figure 1 shows the effects of task (anger recall, reading) and distracter on change scores from

baseline for each of the examined physiological variables.

Discussion

Functional neuroimaging studies have shown that the prefrontal cortex normally exerts tonic inhibitory control on sympathoexcitatory neural circuits; apparently, rumination takes the prefrontal cortex temporarily “off-line,” leading to the disinhibition of these circuits (Brosschot, Van Dijk, & Thayer, 2007). The present study showed the consequences of these processes, parasympathetic withdrawal and relative sympathetic dominance during rumination, thus confirming and extending previous findings of ours (Ottaviani et al., 2008) and others (Key et al., 2008; Ray, Wilhelm, & Gross, 2008; Verkuil et al., 2009).

In contrast with the results of our previous study (Ottaviani et al., 2009), in which we found increased vagal withdrawal in women

compared with men, but consistent with Glynn, Christenfeld, and Gerin (2002) and Suarez, Saab, Llabre, Kuhn, and Zimmerman (2004), there were no sex differences in autonomic activation during the Anger Recall Interview. It has to be noted that in our previous study participants were asked to think silently about a past episode of anger and not to speak it out loud, as in the present experimental protocol and in studies that obtained comparable results (Glynn et al., 2002; Suarez et al., 2004).

In line with what was reported by Bernardi, Wdowczyk-Szulc, Valenti, Castoldi, Passino, Spadacini, and Sleight (2000), reading aloud resulted in sympathetic nervous system activation as shown by increased LF-HRV n.u. and LF/HF-HRV, but not accompanied by a simultaneous reduction of vagal tone. This activation, however, is still lower than the one occurring after the interview Anger Recall task, as shown by direct statistical comparison between the two tasks. According to Marci, Glick, Loh, and Dougherty (2007), feeling angry is generally

accompanied by sympathetic nervous system activation, which is responsible for prolonged rises in BP and consequently increases the risk for hypertension. In the present study, sympathetic activation during the Anger Recall Interview was confirmed by changes in PEP, LF-HRV, and LF/HF-HRV. As already stated, however, the emphasis should not be so much on the fact that participants showed cardiac activation during a stressor such as recalling an episode of anger, but rather on the lack of flexibility that delays or prevents recovery after the stressful environment is no longer present. Under normal conditions, the sympathetic nervous system activity, which is associated with the mobilization of energy, and the parasympathetic nervous system activity, which is associated with vegetative and restorative functions, are in a state of dynamic equilibrium in order to face rapid and continuous environmental demands. If the dominance of one branch of the autonomic nervous system over the other is maintained even after the disap-

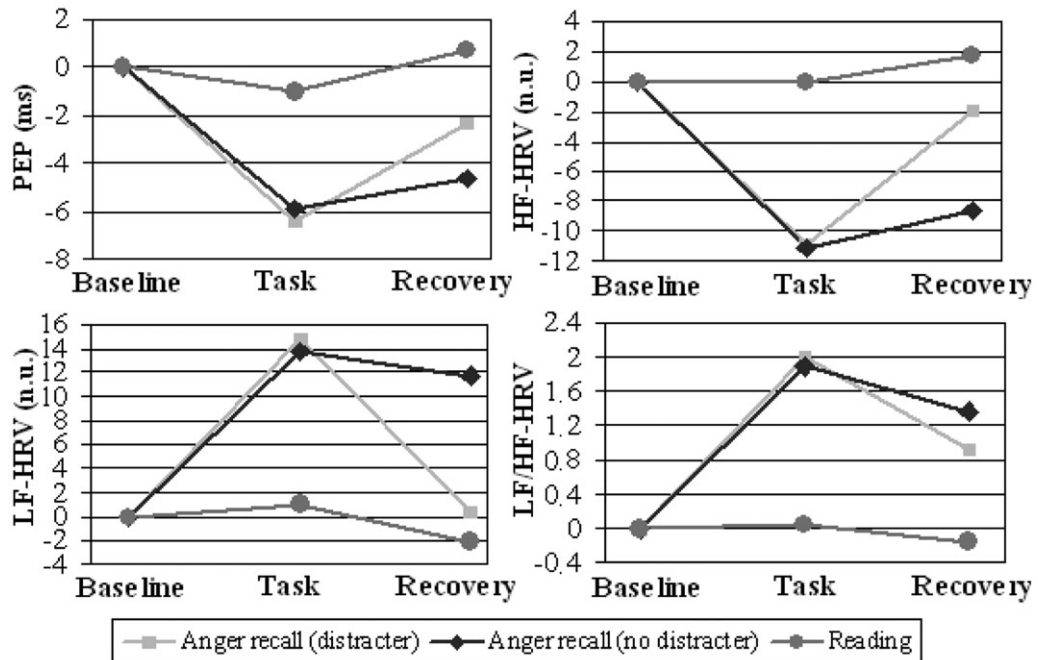


Figure 1 Differences in reactivity and recovery change scores between the anger recall (distracted and nondistracted subgroups) and the reading tasks.

pearance of stressful events, the individual becomes, because of this lack of flexibility and adaptability, vulnerable to disease. Brosschot et al. (2006) suggested that this prolonged physiological activity could be the missing piece in the theoretical model that relates the experience of anger to the development of cardiovascular disease. In accordance, not only talking about an anger episode but also thinking about it was associated with hyperactivation of the sympathetic system and hypoactivation of the parasympathetic system. Specifically, rumination was associated with shorter PEP and higher LF-HRV and LF/HF-HRV on the one hand and lower HF-HRV on the other. This pattern of hypersympathetic and hypoparasympathetic regulation might be best represented as a shift in autonomic balance. Empirically, there is a large body of evidence to suggest that autonomic imbalance is associated with various pathological conditions. In their recent review, Thayer, Yamamoto, and Brosschot (2010) showed that decreased vagal function is an independent risk factor for all-cause mortality, after controlling for other risk factors. The authors assert the need for studies that include indices of both sympathetic and parasympathetic activity as necessary to provide a more complete picture of the role of the autonomic nervous system in health and disease. Present findings emphasize the need for prospective studies to investigate the long-term effects of rumination on health and the urge to implement therapeutic techniques specifically focused on how to cope with perseverative cognitions.

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